IPv4 Vs IPv6

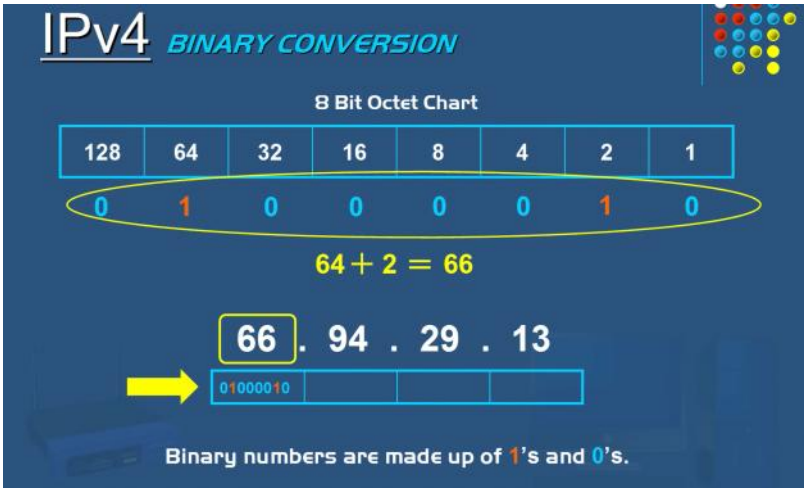
Tuesday, February 24, 2026 5:59 PM

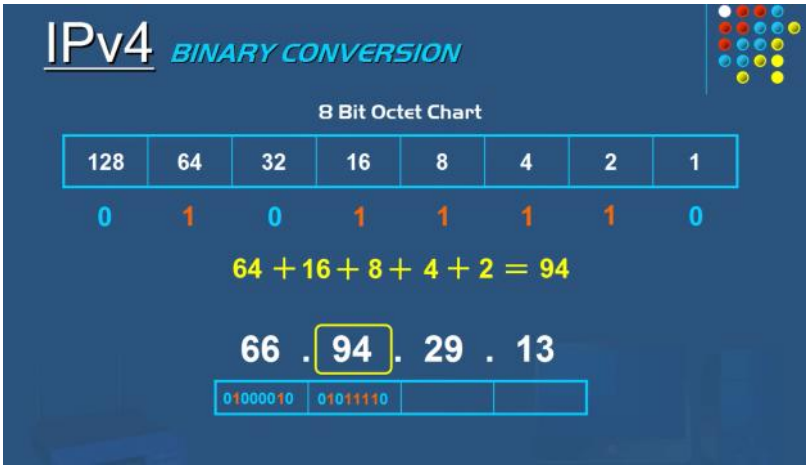
Types of IP Addresses

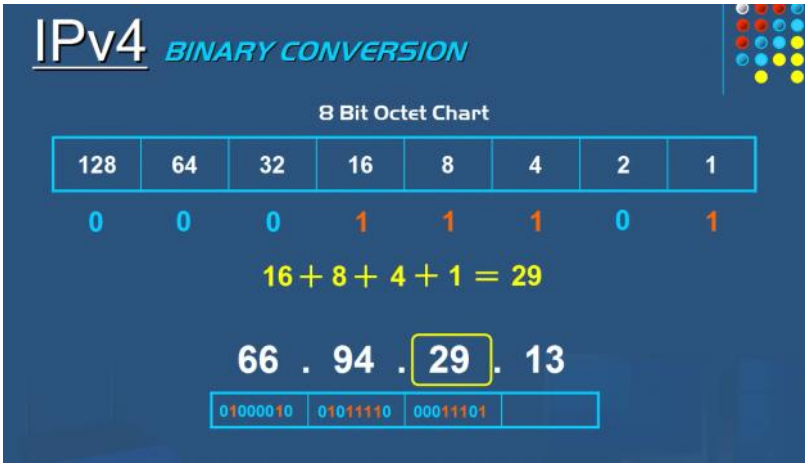
An IP address is a numeric identifier for a device on a network, essential for communication. There are two main types: IP version 4 (IPv4) and IP version 6 (IPv6).

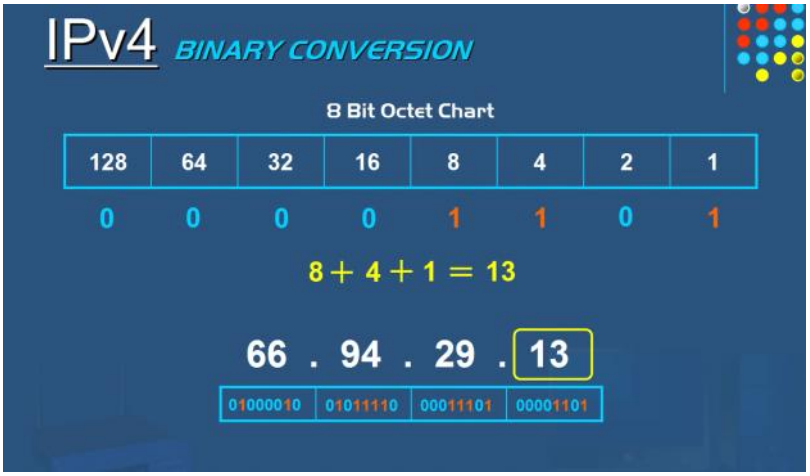
IPv4

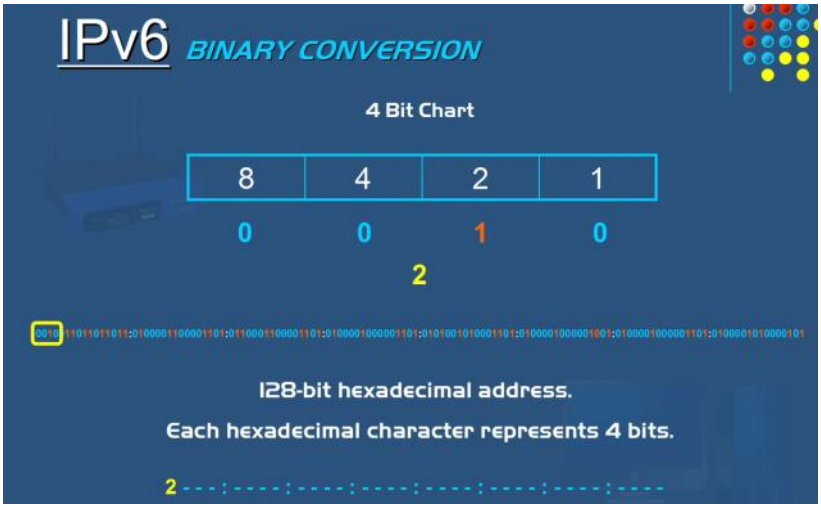
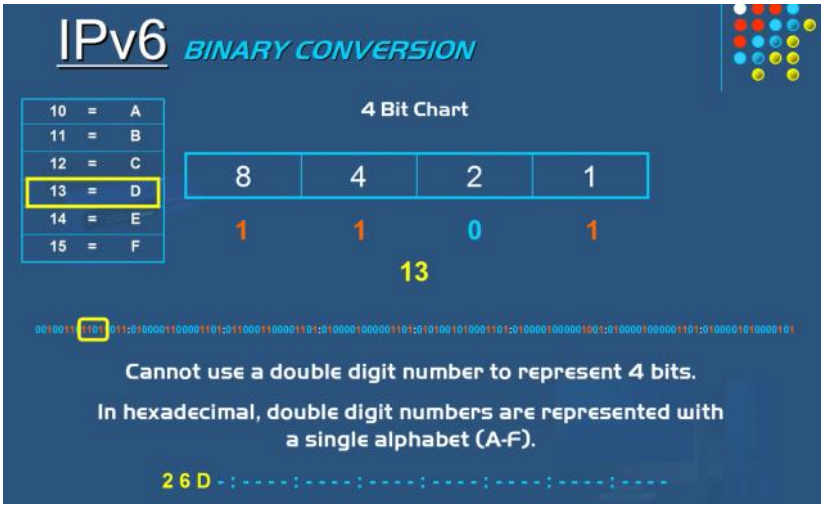
IPv4 is a 32-bit numeric address, written as four numbers separated by periods (octets), ranging from 0 to 255. It can produce over 4 billion unique addresses. Computers read these addresses in a binary format, which uses ones and zeros. The process of converting an IPv4 address to its binary form involves understanding the bit values within each octet.









IPv6

IPv6 is the next generation of IP addresses, a 128-bit hexadecimal address. It uses both numbers and alphabets and can produce an incredibly vast number of unique addresses (340 undecillion). Converting a binary number to an IPv6 hexadecimal address involves representing four bits at a time with a single hexadecimal character, using letters A-F for sums of 10 or higher.